1) $y(n) = a \cdot x(n-T) = a \cdot x(n) * \delta(n-T)$

$$\varphi_{xy}(n) = x(-n) * y(n) = \left(x(-n) * x(n) * \delta(n-T)\right) \cdot a$$

$$= \left(\varphi_{xx}(n) * \delta(n-T)\right) \cdot a$$

$$= a \cdot \varphi_{xx}(n-T)$$

The autocorrelation function $\varphi_{xx}(n)$ has a unique maximum at $n=0$.

$\Rightarrow$ $\varphi_{xx}(n-T)$ has a unique maximum at $n=T$.

2)

$$y_1(n) = x(n) + d_1(n)$$

$$y_2(n) = x(n) + d_2(n)$$

$x(n)$ = wanted signal
$d_1(n)$ = noise / distortion of mic 1
$d_2(n)$ = -"- mic 2

output of delay and sum beamformer $y(n) = \left(y_1(n) + y_2(n)\right)\frac{1}{2}$

$$SNR = 10 \log_{10} \frac{\sum_n x^2(n)}{\sum_n \left(y(n) - x(n)\right)^2}$$

$$= 10 \log_{10} \frac{\sum_n x^2(n)}{\sum_n \left(x(n) + \frac{1}{2} d_1(n) + \frac{1}{2} d_2(n) - x(n)\right)^2}$$

$$= 10 \log_{10} \frac{\sum_n x^2(n)}{\frac{1}{4} \sum_n \left(d_1(n) + d_2(n)\right)^2}$$

$$= 10 \log_{10} \frac{4 \sum_n x^2(n)}{\sum_n d_1^2(n) + 2 d_1(n) d_2(n) + d_2^2(n)}$$

$$= 10 \log_{10} \frac{4 \sum_n x^2(n)}{\sum_n d_1^2(n) + 2 \underbrace{\sum_n d_1(n) d_2(n)}_{\approx 0 \text{ due to uncorrelated noise}} + \sum_n d_2^2(n)}$$

$$= 10 \log_{10} \frac{\cancel{4}^{\,2} \sum_n x^2(n)}{\cancel{2} \sum_n d_1^2(n)}$$

assuming that noise energy is equal for both microphones:
$\sum_n d_1^2(n) \approx \sum_n d_2^2(n)$

SNR without delay and sum beamformer:

$$SNR = 10 \log_{10} \frac{\sum_n x^2(n)}{\sum_n d_1^2(n)} \approx 10 \log_{10} \frac{\sum_n x^2(n)}{\sum_n d_2^2(n)}$$

Gain due to beamformer: $G = 10 \log_{10} \frac{2 \sum_n x^2(n)}{\sum_n d_1^2(n)} - 10 \log_{10} \frac{\sum_n x^2(n)}{\sum_n d_1^2(n)}$

$$= 10 \log_{10} \left(\frac{2 \sum_n x^2(n)}{\sum_n d_1^2(n)} \cdot \frac{\sum_n d_1^2(n)}{\sum_n x^2(n)} \right)$$

$$= 10 \log_{10} 2$$

$$= 3{,}0 \text{ dB}$$

3) N Microfones

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(n) + d_k(n)$$

$$\text{Noiselevel} = \left(y(n) - x(n)\right)^2 = \left(\frac{1}{N} \sum_{k=0}^{N-1} d_k(n)\right)^2 = \frac{1}{N^2}\left(\sum_{k=0}^{N-1} d_k^2(n) + \underbrace{2 d_0(n) d_1(n) + 2 d_0(n) d_2(n) + \dots}_{=0\text{, due to uncorrelation}}\right)$$

$$= \frac{1}{N^2} \sum_{k=0}^{N-1} d_0^2(n)$$ assuming equal noise level for each microfone

$$= \frac{1}{N^2} \cdot N \cdot d_0^2(n)$$

$$= \frac{1}{N} d_0^2(n)$$

$$\text{SNR Gain} = 10 \log_{10} \frac{\sum_n x^2(n)}{\frac{1}{N} d_0^2(n)} - 10 \log_{10} \frac{\sum_n x^2(n)}{d_0^2(n)}$$

$$= 10 \log_{10} N$$

4) $$y(n) = a\, x(n-T) = x(n) * \underbrace{a\,\delta(n-T)}_{\text{channel model}}$$

$$\varphi_{xy}(n) = x(-n) * y(n) = x(-n) * x(n) * a\,\delta(n-T) = \varphi_{xx}(n) * a\,\delta(n-T) = a\,\varphi_{xx}(n-T)$$

$\varphi_{xx}(n)$ has a unique maximum for $n=0$.

$\Rightarrow$ $\varphi_{xy}(n)$ has a unique maximum for $n=T$.

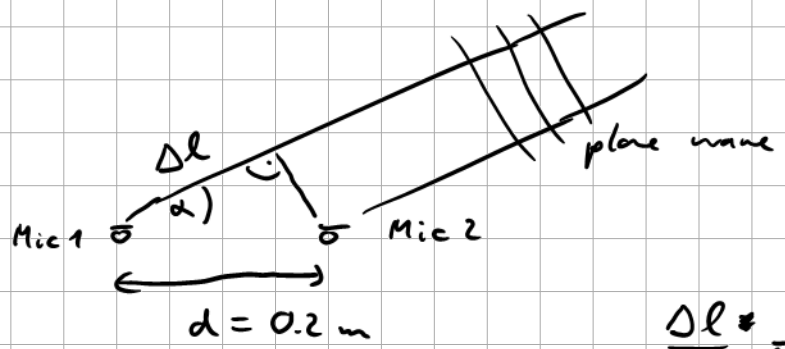
$\Rightarrow$ $T =$ numpy.argmax (phi_xy)

$$\varphi_{xy}(T) = a\,\varphi_{xx}(0)$$

$$\Rightarrow \quad a = \frac{\varphi_{xy}(T)}{\varphi_{xx}(0)}$$

$$\Rightarrow \quad a = \frac{\text{phi_xy}[T]}{\text{phi_xx}[0]}$$

5)



$$\frac{\Delta l}{d} = \cos\alpha \Rightarrow \Delta l = d \cdot \cos\alpha$$

time difference in seconds $= \Delta t = \frac{\Delta l}{c} = \frac{d \cdot \cos\alpha}{c}$

time difference in samples $= \Delta n = \Delta t \cdot r = \frac{d \cdot \cos\alpha}{c} \cdot r$

$$\Delta n = \frac{0.2\,m \cdot \cos 30°}{343\,\frac{m}{s}} \cdot 48000\,Hz = 24{,}2$$

$\Rightarrow$ Delay of 24 samples necessary!

6) <u>uncorrelated:</u> $\sum_n x_1(n) \cdot x_2(n) \approx 0$

$$y(n) = x_1(n) + x_2(n)$$

$$E_y = \sum_n y^2(n) = \sum_n \left(x_1(n) + x_2(n)\right)^2$$

$$= \sum_n x_1^2(n) + 2 \underbrace{\sum_n x_1(n) x_2(n)}_{\approx 0} + \sum_n x_2^2(n)$$

$$= \sum_n x_1^2(n) + \sum_n x_2^2(n)$$

$$= E_{x_1} + E_{x_2}$$

<u>correlated</u> $x_1(n) \approx c \cdot x_2(n) \qquad c \in \mathbb{R}$

$$y(n) = x_1(n) + x_2(n)$$

$$= (c+1)\, x_2(n)$$

$$E_y = \sum_n y^2(n) = \sum_n (c+1)^2 x_2^2(n)$$

$$= (c+1)^2 \sum_n x_2^2(n)$$

$$= (c+1)^2 \quad E_{x_2}$$

$$E_{x_2} = \sum_n x_2^2(n)$$

$$E_{x_1} = \sum_n x_1^2(n) = \sum_n c^2 x_2^2(n) = c^2 E_{x_2}$$

$$E_y = (c^2 + 2c + 1) E_{x_2}$$

$$= E_{x_1} + \underbrace{2c \cdot E_{x_2}}_{\text{additional energy due to correlation}} + E_{x_2}$$